electron/workspace-store.cjs

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Overview

The workspace store now serializes session writes, uses atomic file writes, and simplifies the `upsertWorkspace` flow. The public API remains unchanged.

Key changes

- **Session write queue** – a global `sessionWriteChain` (L25-32) and `withSessionWriteLock(task)` (L27-31) serialize updates to the session file.
- **Atomic writes** – `writeJsonAtomic` (L33-63) writes to a temporary file, renames it, and falls back to `copyFile` on rename errors (lines 47-50, 55-58).
- **`writeJson`** now delegates to `writeJsonAtomic` (L64-67).
- **`saveSession`** wraps its write in the lock (L101-102).
- **`upsertWorkspace`** signature changed to `upsertWorkspace(userData, workspace, opts = {})` (L110-111). It writes the workspace meta first (`writeJson(path.join(workspace.dataDir, "meta.json"), workspace)` – L112-113), then updates the session inside the lock (L116-129). The old implementation (L74-86) was removed.
- **Error handling** – `writeJsonAtomic` now checks `fssync.existsSync(tmp)` before attempting `copyFile` and `unlink` (lines 47-50).

Impact

- **Correctness** – atomic writes and the write lock prevent partial or corrupted session files, especially under concurrent operations.
- **Maintainability** – centralizing write logic reduces duplication and clarifies persistence flow.
- **Compatibility** – exported API unchanged; callers use the same functions.

Risks & follow-ups

- Verify that `withSessionWriteLock` propagates rejections and that `sessionWriteChain` resets after a failure.
- Run concurrent stress tests on `saveSession` and `upsertWorkspace` to ensure no data loss.
- Confirm the rename-fallback logic works on Windows and other filesystems where `rename` may fail.
- Ensure `writeJsonAtomic` still appends a newline (`\n`) to JSON files, preserving the original formatting.